



# Databases and Applications

## Unit III

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Computer Applications

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BASE Principles, CAP Theorem, RDBMS vs NoSQL

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## BASE Principles – NoSQL Reliability Model

### ► Why BASE emerged ?

- Traditional ACID models become restrictive in distributed clusters
- Large-scale web systems need availability even during partial failures
- BASE trades strict correctness for operational resilience and performance

### ► Detailed BASE Semantics

- **Basically Available:** System continues operations even if some nodes are unreachable
- **Soft-State:** Replicas may temporarily hold different versions of data
- **Eventual Consistency:** Background reconciliation ensures replicas converge



### ► What is BASE?

- **Basically Available** – system remains available even during failures
- **Soft-state** – data may be temporarily inconsistent
- **Eventual consistency** – data becomes consistent over time

### ► Why BASE is used in NoSQL

- Designed for distributed, large-scale environments
- Prioritises performance and scalability over strict consistency
- Suitable for real-time web, mobile, IoT applications



### ► **ACID in RDBMS (PostgreSQL / Oracle)**

- Enforces correct, predictable transactions under a strict schema
- Ensures durability through WAL (Write-Ahead Logging) and checkpointing
- Suitable for banking, ERP, inventory, and mission-critical records

### ► **BASE in NoSQL (MongoDB Documentation)**

- Prioritises throughput and latency over immediate consistency
- Replica sets sync asynchronously for speed
- Enables flexible schema evolution without table redesign



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## CAP Theorem – Distributed Systems

### ► CAP Theorem States:

- A distributed system can guarantee only **two** of:
- Consistency (C)
- Availability (A)
- Partition Tolerance (P)

### ► Real-world Interpretation

- Network failures force trade-offs between C and A
- No database can provide all three at all times
- NoSQL DBs balance A+P; RDBMS balances C+P



### ► The CAP Trade-off (Academic View)

- **Consistency (C):** Every read returns the most recent committed write
- **Availability (A):** Every request receives a non-error response
- **Partition Tolerance (P):** System continues despite message loss or network splits

### ► Implications

- In a real distributed system, network partitions are unavoidable
- Systems must choose **CA**, **AP**, or **CP** depending on business need
- No system can simultaneously guarantee C, A, and P in the presence of failures



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## RDBMS vs NoSQL – When to Use What?



### ► RDBMS (MySQL, PostgreSQL)

- Structured schema with fixed tables
- Strong ACID guarantees
- Best for finance, transactions, ERP systems
- Fixed schema with strong typing and relational integrity constraints
- Vertical scaling (scale-up) using single-node performance
- Optimised for OLTP workloads and structured datasets

### ► NoSQL (MongoDB, Cassandra)

- Flexible schema for semi-structured data
- Scales horizontally across clusters
- Schema-flexible documents enabling heterogeneous records
- Horizontal scaling (scale-out) through sharding and replica sets



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